DDI Implementation Projects at SSJDA Ongoing Projects and a Future Task

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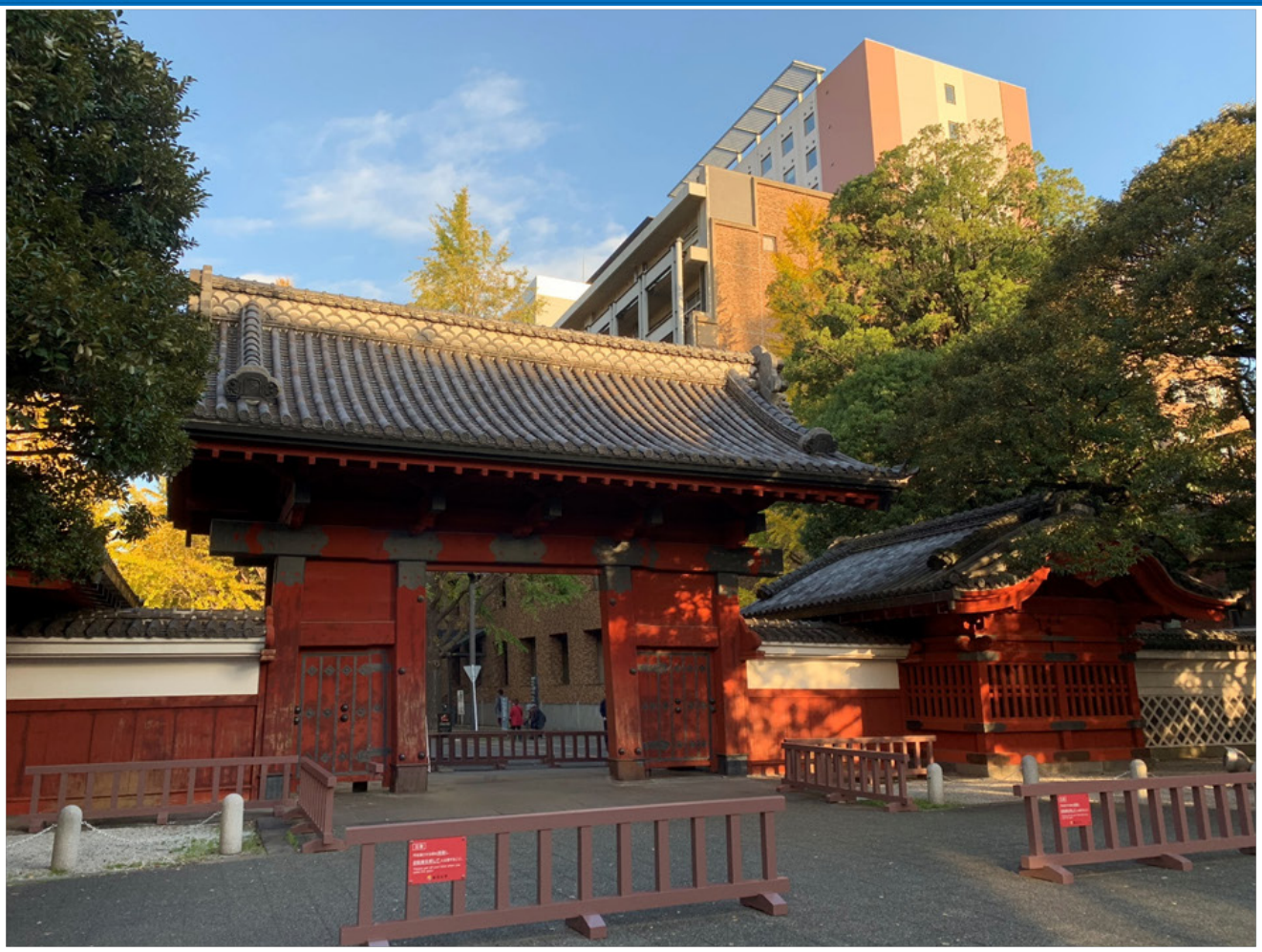

Social Science Japan Data Archive (SSJDA) has developed DDI implementation projects. We present two ongoing projects and a future task in SSJDA. The first implementation project is the Nesstar system operation since 2012. Some datasets which are provided from SSJDA are available in the Nesstar system and the number of access and users are increasing. Since mutual link with SSJDA's website is also started, cooperation between services is working more effectively.

The second project is development of the software Easy DDI Organizer (EDO). EDO is an open source tool for editing and managing a metadata along with data lifecycle.

Our future task is to provide users with tools to encourage users to understand and use social research. Panel surveys usually have complex structures that impede analysis. At the same time, I would like to integrate these multiple projects and improve the process of data archiving work. Discussion of these projects and future issues.

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Until now and about the current DDI project

We will introduce the DDI introduction projects that SSJDA has been conducting so far. I would like to share the points that we are addressing in these projects and our future goals. There are two major DDI projects that we have been working on. One is the Nesstar installation project, and the other is the creation of the original DDI Editor.

Nesstar project

The project introduces the Nesstar system provided by NSD in 2012. Nesstar is an application with on-line analytics and DDI-based metadata management mechanisms. An online analysis application was being used previously; however, as the lifecycle of the application (SPSS WebApp) had ended, we were on the lookout for a new online analysis software. In 2014, full-scale operations were implemented.

https://nesstar.iss.u-tokyo.ac.jp/webview/

This project has contributed to bringing the concept of DDI to our archive for trial. The DDI person in charge reconstructs the metadata from the data published in SSJDA-Direct and uploads it to Nesstar together with the dataset.

We are currently operating Nesstar as we periodically add new datasets into the system. Currently, more than 100 data can be used and English data (Japanese General Social Surveys) is also being handled. In this project, there is a problem of dual management of SSJDA-Direct metadata and Nesstar's metadata. Based on future maintainability, I think that it is necessary to change the operation method somewhere. Furthermore, Nesstar's powerful functions, such as codebook generation function, are currently not being used.

About DDI Editor <Easy DDI Organizer> EDO Project.

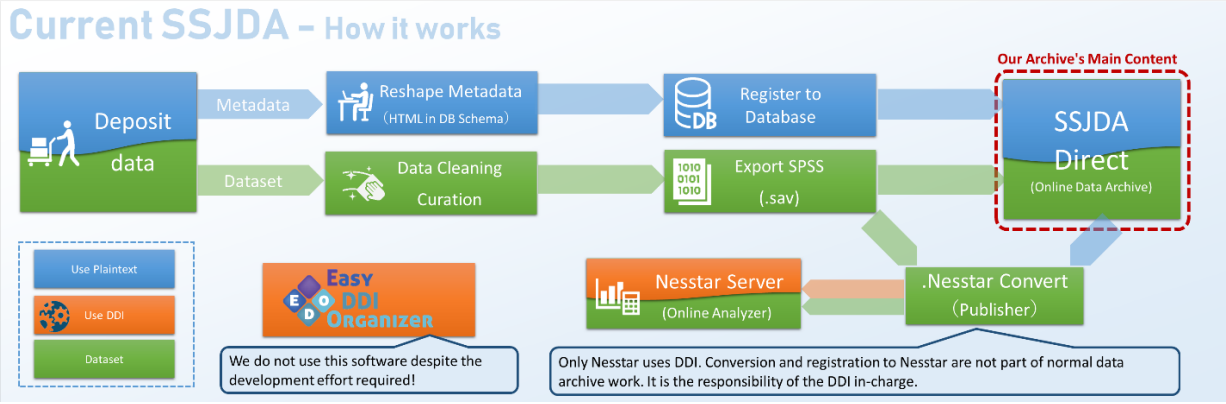
In 2010, we decided to develop an editor that can create DDI data. We introduced the concept of data lifecycle in the editor, and it was designed for the purpose of "generating metadata from the beginning of the survey to the end of the survey". We were using .NetFramework4 and WPF, which was a new technology at that time, in the editor (Ribbon UI was introduced by novelty).

https://csrda.iss.u-tokyo.ac.jp/en/international/ddi/edo/

After that, we have continued development such as the addition of a function to read SPSS data and generate a codebook. From this year onwards, we are also planning some new developments.

The problem with this project is its low usage frequency. You can check in the figure that this tool is not related to the process of data archiving. In the future, we hope that EDO will function in a way that is more aligned with our existing data archiving process.

About current SSJDA work

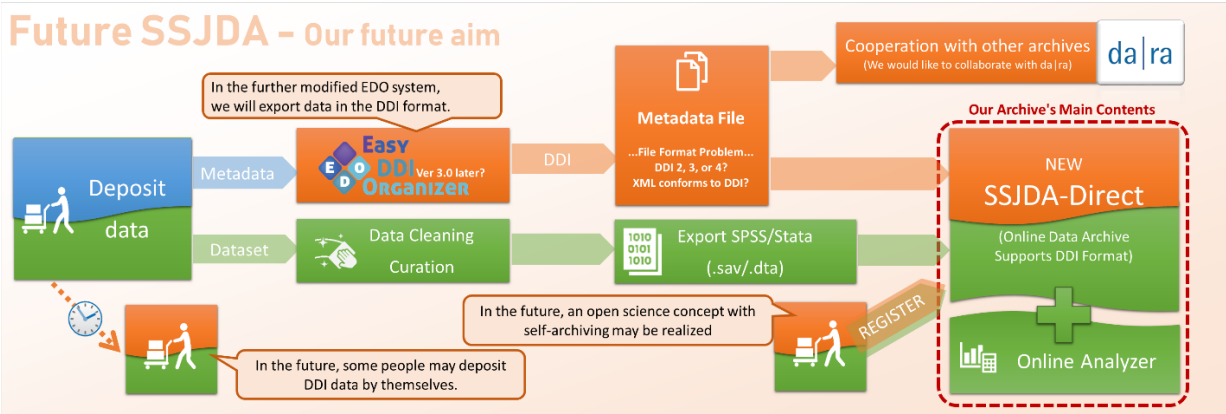


Based on these two projects, the workflow of our current SSJDA is expressed in the upper diagram.

Our main task is to receive data-/metadata-related information from the depositary, register it in the SSJDA-Direct system and make it available to users. In fact, this flow does not contain any DDI elements at all.

Here, blue denotes plain text, green denotes dataset, and DDI-like data is expressed in orange. As you can see, metadata information is mostly represented in blue.

DDI is used at the timing of registration to the Nesstar system. At this time, the person in charge creates Nesstar data based on the metadata information on SSJDA-Direct. We are currently dealing with Nesstar as an online analytical system, the metadata is in a dual management state, and the metadata of plain text (HTML)-based SSJDA-Direct is positive.



In this situation, it is difficult to publicly say that DDI is used in our business. Our goal is to place the concept of DDI at the core of the current workflow. This vision is expressed in the lower diagram saying "Future SSJDA."

We would like to start recording on DDI data using EDO at the time of data deposition. Then, we will not confine the Word file or text file in DB Schema-based HTML. As a result, this file will form the core of the metadata within SSJDA. Furthermore, the SSJDA-Direct system will also support DDI-based information.

Need to Improve EDO

To build this new system, refurbishing of EDO is necessary, which can generate the DDI compliant metadata file. The staff currently responsible for "Reshape Metadata" directly registers the SSJDA-Direct system via the browser. For that purpose, I am planning to implement the same user interface as the current system on the EDO system.

Choosing the file format while saving the file is a critical issue. The excellent format provided by DDI is somewhat fluctuating and has some technical difficulties. We are considering original XML schemar that can output DDI format files. By placing this XML at the core, I am planning to correspond not only to the DDI format of each version but also to a proprietary format required for cooperation with other archives. (The biggest problem is how to assign a field. First, we need to consider the outer mechanism.)

Until now, the EDO system was made with awareness of the data lifecycle. Future updates does not change this policy, and we believe that EDO will be more in line with the current situation of data operation in Japan. By continuing to develop this tool, I hope that someone will deposit DDI data on his own (We may need to do enlightenment activities).

About New SSJDA-Direct

We will also renovate the SSJDA-Direct system that provides metadata browsing and downloading of data, and is the main content of our SSJDA. In the current major updates of the SSJDA-Direct system, we are trying to realize metadata management in DDI format by matching the information granularity with the DDI format. Mismatch of information granularity is a considerable problem that causes phenomena such as putting important information into “Abstract”. We must solve these problems.

We are not aiming for a revolutionary evolution but rather considering a mechanism to increase metadata on DDI. Therefore, we consider parallelly performing existing metadata registration on DB Schema-based HTML and registration by DDI. Simultaneously, making changes to existing items of metadata so as to conform to the DDI format is also necessary.

Metadata in the SSJDA-Direct system is now under dual management with NESSTAR. For unified management of metadata, we will update the SSJDA-Direct to handle the DDI format. Currently, our main goal is to be able to freely download metadata conforming to DDI format from the SSJDA-Direct webpage.

Furthermore, in accordance with the concept of open science, an introduction to self-archiving is also under consideration in SSJDA, but it is uncertain whether it will be implemented. In the future, if we create a system that allows depositors to register their own metadata on the DDI format, it will be possible for users to download complete DDI metadata without undertaking an extra conversion operation.

About cooperation with other archives

We think that it is necessary to declare "Hello World" as soon as possible. For cooperation using DDI, coding the first step is necessary. International cooperation will be carried out more smoothly by implementing a full-scale introduction of the DDI concept.

We will not keep a wait-and-see attitude. We will search methods for cooperating with other archives while manually converting metadata. (We need a little more time until “harvesting” becomes possible.)

Japan seems to be considering launching an organization that grants DOI. However, elsewhere in East Asia, since Taiwan's SRDA precedes us in metadata cooperation using "da|ra", SSJDA is also trying to keep pace.

Q: How many years will it take to realize this plan?

A: I think that several years will be needed until the concept of DDI has been properly incorporated, (and not a barebone implementation) into our work. All projects will simultaneously evolve in small increments.

Vielen Dank für das Lesen bis zum Ende!